Gaggiuino Gen 2 User Manual: Gaggia Classic Pro

Blackpill (STM32F411) controller with a 4.3" Nextion touchscreen.
Prepared 3 November 2026.

> *Built from: the official Gen 2 user manual (gaggiuino.github.io, page dated 03/11/2026), this machine's firmware source (build c1663ac plus local changes), the 4.3" display project file (Nextion 26JUN2024), and photos of the screens. The official text was rewritten in my own words. Where this manual describes behaviour that comes only from reading the code, or from changes made to this particular build, it says so.*

1. About this manual

Gaggiuino is an open-source control system that replaces the stock thermostat logic of a Gaggia Classic or Classic Pro. It adds precise boiler temperature control, pressure and flow profiling, shot timing and graphing, stop-on-weight, and an automatic descale routine, while keeping the original three switches.

The official manual was written around the smaller screen. Your 4.3" display shows the same features in a different layout:

- The brew pages have a tab strip reading **PI**, **Soak**, **Trans**, **Profile**, **Manual**, **More**. The official manual calls the Trans page "Advanced".
- The Settings pages have **Temp** and **System** tabs, a **Build ver.** and **Nextion ver.** line at the bottom of the System page, and a **Standby Timer** section on the More page.

If something on your screen does not match this manual, the screen is correct. Please tell the person maintaining the build.

2. Safety

- **Mains voltage is present inside the machine.** Unplug the machine before opening it or touching any wiring. Heater and pump wiring carries lethal voltage.
- **Steam and hot water cause burns.** The boiler runs well above 100°C in steam mode. Keep the steam valve and wand clear, and never block the group head.
- **Never leave steam mode switched on unattended.** The controller shuts the heaters down after about 15 minutes of forgotten steam, but do not rely on that.
- **Descaling uses hot, acidic solution.** Follow section 10 exactly, and rinse with clean water afterwards.
- **Modifying the firmware is at your own risk.** If the temperature sensor fails, the controller is designed to switch off the pump and heaters, but only if it is running the correct firmware.

3. What is in this machine

Based on the Gaggiuino Lego build schematic for this machine:

- **Gaggia Classic Pro:** Boiler, pump, group, three-way valve, steam wand
- **Blackpill STM32F411 on a Nano expansion board:** The controller; runs the Gaggiuino firmware
- **4.3" Nextion touchscreen:** The user interface
- **MAX6675 thermocouple amplifier:** Reads boiler temperature
- **Pressure transducer with ADS1015 converter:** Reads system pressure
- **Pump dimmer:** Lets the controller vary pump power for pressure and flow control
- **Solid-state relay:** Switches the boiler heater
- **Two HX711 load-cell boards:** Drip-tray scales (optional)
- **ToFnLED board:** Measures tank water level and lights the tank; connects on Blackpill pins B6 and B7

The distance sensor and LED are enabled by build settings (section 16). Without the hardware scales, the controller estimates weight from pump flow instead ("predictive scales").

4. Switches and controls

- **Brew switch:** starts and stops a shot. In Flush mode it runs a flush, and in Descale mode it runs the descale program.
- **Steam switch:** puts the machine in steam mode (target about 155°C by default).
- **Hot water switch:** runs hot water mode.
- **Touchscreen:** everything else, including profiles, settings and cleaning programs.

With hardware scales fitted, the scales tare automatically when a shot starts. Switching the brew switch on also wakes the screen if it was asleep.

5. Starting up



1. **Switch on.** The splash screen shows a progress bar and a build number in the corner. If the number is missing, the display and the controller are not communicating.
2. **Boiler fill.** About 3 seconds after start, the pump runs briefly to fill the boiler. The screen shows "Filling boiler!". It stops when the boiler is full, or after 8 seconds at most. It is skipped if the boiler is already hotter than 85°C. If the brew switch is on during startup, the screen shows "Brew Switch ON!" and nothing starts until you turn it off.
3. **Warm up.** Allow 10 to 15 minutes for the whole machine to come up to temperature. The uptime counter on the home screen shows how long it has been on. If the Warmup setting is on, a "Warming Up" message flashes for 15 minutes after power-on.
4. **Water level.** If the tank drops below 10%, the screen shows "Fill the water tank!".

After startup the machine heats to the active profile's temperature.

6. The home screen



The home screen shows:

- Menu tabs: Home, Brew, Clean and Settings.
- Profile list: five profile slots. Tap one to select it.
- Target temperature: normally the brew temperature of the selected profile (about 93°C), and the steam temperature when steam is on.
- Water temperature: the large ring in the middle.
- Pressure: the current system pressure in bar.
- Tank water level: a bar from 0 to 100% (needs the ToFnLED).
- System uptime: counts from power-on.
- Scales and brew ratio: the weight and the shot target (see below).

Pressure release ("dropping beats")

When the machine is idle and there is still pressure above 0.7 bar in the system, the controller releases it through the group head. It counts down on screen first, shows "Releasing pressure!", and equalises the system so the next shot starts from the same conditions. The check runs about every 30 seconds.

Selecting and naming profiles

Tap a profile to select it. Long-press a profile to open the Custom Profile Override screen:



- **Profile name box:** type a new name.
- **OK:** saves the name and makes this profile the one loaded at start-up. You do not have to rename it to make it the default.
- **Load default:** restores that slot to the factory profile and returns home without changing the start-up profile.

To build a new profile, select a slot, edit the Brew pages (section 8), save, then long-press the profile on the home screen and press OK.

Brew ratio and stop on weight

Open the brew ratio area on the home screen to set:



- **Stop on weight:** on or off. When on, the shot stops by itself at the target weight.
- **Coffee dose:** the weight of ground coffee in grams.
- **Brew ratio:** presets 1:1, 1:2, 1:3 and 1:4. The target output is dose times ratio.
- **Shot weight override:** type a target weight in grams to ignore the dose and ratio. Set 0 to turn the override off.



Stop on weight works with hardware scales and with predictive scales. The controller looks about half a second ahead, so the shot stops slightly before the target to allow for water still falling into the cup.

Scales

Tap the weight box to see the weight in grams. Tare zeroes it. Hardware scales tare by themselves when a shot starts.

7. Making a shot

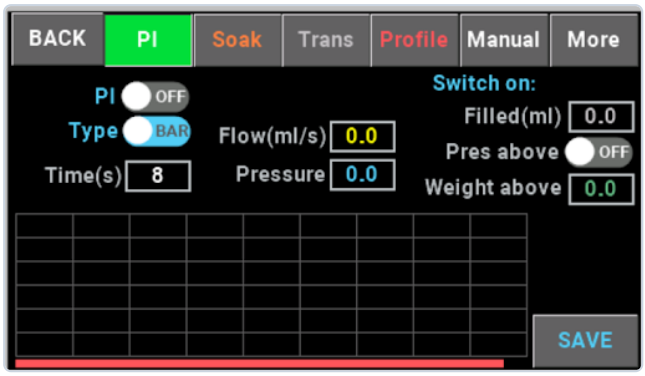
1. Warm up the machine and select a profile. Check the target and the stop-on-weight settings.
2. Grind, dose and tamp, lock the portafilter in, and put the cup (on the scales, if fitted) under the spout.
3. Turn the brew switch on. The screen changes to the brew graph and the timer starts. The scales tare, and the controller runs the profile phases in order: optional basket fill, pre-infusion, soak, transition, then the main profile.
4. The shot ends when the target weight is reached, or when you turn the brew switch off.
5. Turn the brew switch off. The screen returns home after 10 seconds if Home On Finish is on.

Predictive scales: if you have no hardware scales, press the Reset weight button on the graph page when the first drops reach the cup. This lines the estimate up with the real start of the pour.

8. Brew pages and profiles

A profile is a recipe made of up to four phases, which run in this order: pre-infusion, soak, transition, profile. Each can be switched on or off. Every phase has a target (a pressure in bar or a flow in mL/s), and ends when one of its exit conditions is met. Use the Preview button on each page to see the curve before you brew.

8.1 PI (pre-infusion)

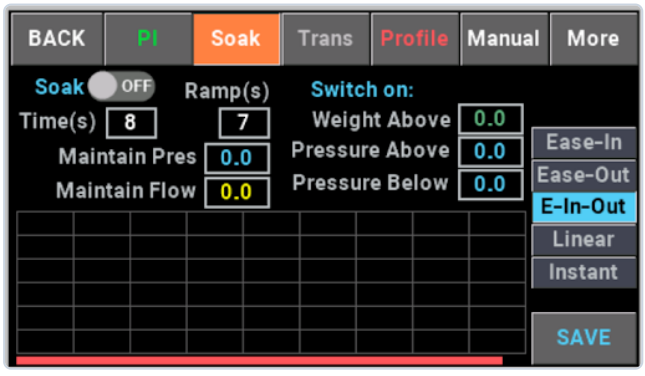


Fills the basket gently so the puck saturates evenly, which reduces channelling and helps lighter roasts.

- PI on/off.
- Flow/Bar: whether the target is a flow or a pressure.
- Time, Flow, Pressure: how long the phase lasts and its target.
- Filled (mL): moves on once this much water has been pumped.
- Pressure above: when on, reaching the pressure ends the phase. When off, the pressure acts only as a ceiling.
- Weight above (g): moves on once this much coffee is in the cup.

Example: 8 mL/s target, 20 s time, 4.5 bar, 100 mL filled, 5 g weight, pressure above off. The machine aims for 8 mL/s but never exceeds 4.5 bar. A fine grind may only manage about 4 mL/s. The phase ends when 100 mL has been pumped, 5 g is in the cup, or 20 s has passed, whichever comes first.

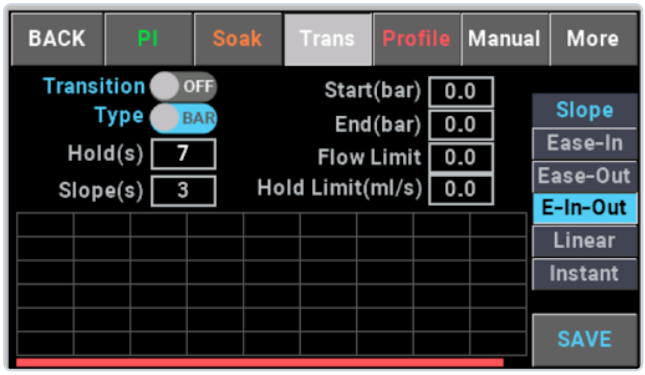
8.2 Soak



Stops the pump so pressure from the pre-infusion can spread through the puck. This can help release trapped CO_2 and ease extraction, and may not suit dark roasts.

- Soak on/off and Time.
- Maintain pressure and Maintain flow: if only one is above zero, the machine holds that target. If both are above zero, it treats them as a pressure target with a flow limit. If both are 0, the pump stays off and pressure falls gradually.
- Exit conditions: weight above, pressure above, or pressure below.
- Slope time and slope type: how long to take to reach the target, and the shape of the curve (see Easing, section 8.6).

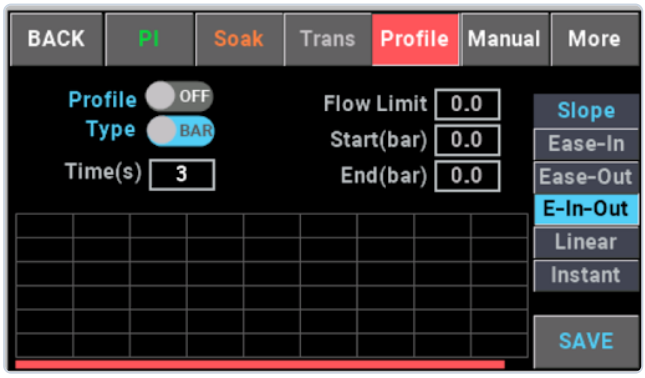
8.3 Trans (transition)



On the official manual this is the Advanced page. It sits between soak and the main profile, for when you want to hold a level before the profile begins.

- On/off, Flow/Bar, Start, End.
- Hold (s): how long to keep the Start value before ramping.
- Slope time, slope type.
- Limit and Hold limit: the ceiling in bar or mL/s while ramping and while holding.

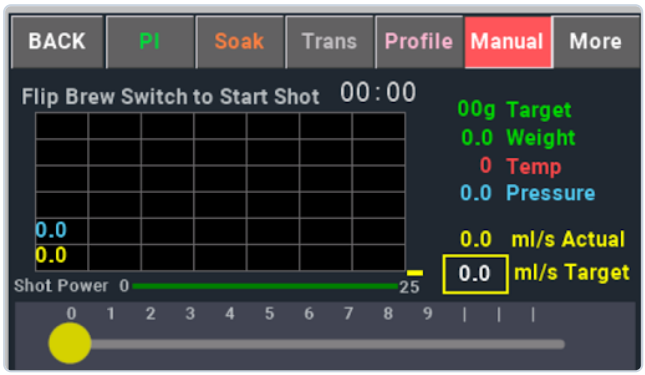
8.4 Profile



The main extraction phase.

- On/off, Flow/Bar.
- Start and End: where the phase begins and ends, rising or falling, in bar or mL/s.
- Slope time and slope type.
- Limit: the machine will not exceed this value.
- Preview.

8.5 Manual



Flow is set by a slider. Move the slider to the target flow in mL/s and turn the brew switch on. The page shows temperature, flow (g/s), pressure, weight and the shot timer.

8.6 Easing

- Ease-in: slow start, fast finish.
- Ease-out: fast start, slow finish.
- Ease-in-out: slow start and finish.
- Linear: steady rate of change.
- Instant: jumps straight to the target.

8.7 More



- **Home On Finish:** returns to the home screen 10 seconds after a shot ends.
- **Brew Delta:** while a shot runs, the boiler target rises in step with the flow so the heater keeps up with the cold water coming in. The boiler can go above 100°C during a shot.
- **Basket Prefill:** before the profile starts, releases pressure and fills the basket at about 7 mL/s until pressure settles at 0.1 bar or more.
- **Standby Timer, Seconds, Idle Temp:** see section 11.

8.8 The five factory profiles

- **IUIUIU Classic (93°C):** All-rounder: flow-driven pre-infusion for an even puck, a falling-pressure soak, then a pressure profile limited in flow. Needs some tweaking per bean.
- **Londinium (92°C):** Imitates a Londinium R lever machine; smooth, with body. Suits most beans and roasts.
- **Adaptive (95°C):** Copes with different grind sizes by setting the extraction flow relative to the peak-pressure flow, with a pressurised bloom. Best for light roasts; suggested ratio 1:2 to 1:2.5 and 25 to 45 seconds. For darker roasts try lower temperatures and a shorter bloom.
- **Filter 2.1 (89°C):** Filter-style coffee from the espresso machine, with a paper filter and a puck screen. Pull to about 5:1 and dilute with roughly 225 to 250 g of water.
- **Blooming espresso (93°C):** Flow-controlled pre-infusion and a long bloom that saturates the puck; aims for pour-over-like flavour and less sourness in light roasts.

The temperatures above are the factory values in this firmware. They can be changed per profile on the Temp page.

9. Steam and hot water

Steam. Turn the steam switch on. The target changes to the steam temperature (155°C by default) and the screen shows "Ready to steam!" when it is hot. Open the steam wand valve to release pressure if the temperature stops rising.

Safety behaviour built into the firmware:

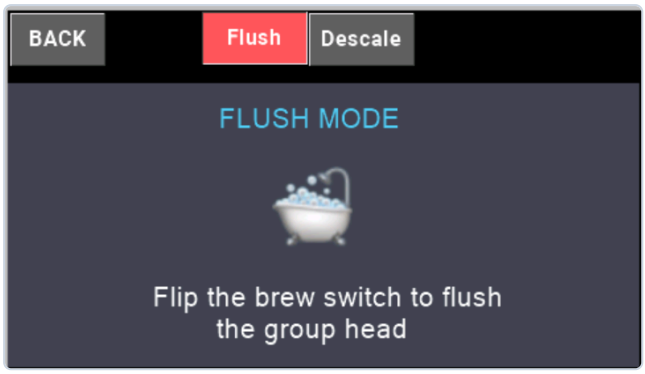
- The heater and steam relays switch off if pressure reaches about 11 bar or the temperature goes over the steam target.
- While steaming, the pump adds water automatically when pressure falls below about 2 bar ("DreamSteam"), to keep the boiler topped up.
- If steam is left on with pressure sitting above about 3 bar for 15 minutes, the heaters shut down and the screen shows "TURN STEAM OFF NOW!".

Hot water. Turn the hot water switch on. The pump runs and the heater is kept below about 105°C.

10. Clean: flush, backflush and descale

Open the Clean tab. It has two parts: Flush and Descale.

Flush



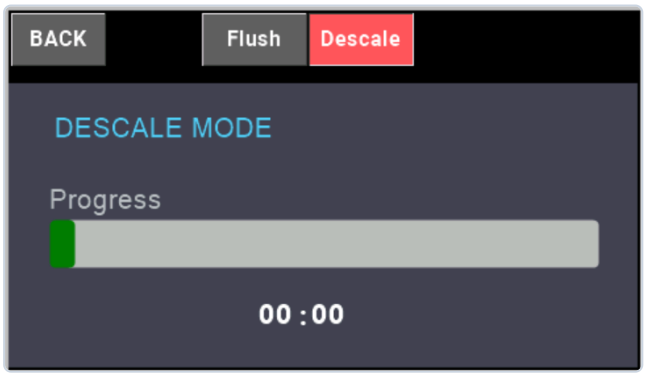
After-shot flush: remove the portafilter and puck, turn the brew switch on for a couple of seconds, then off.

Back-flush through the three-way valve:

1. Fit a blind basket in the portafilter.
2. Optional: remove the shower screen and put cleaning powder in the blind basket.
3. Lock the portafilter in.
4. Turn the brew switch on and let the machine cycle several times.
5. Turn the brew switch off.

In the firmware, the cycle waits until pressure passes 5 bar after about 7 seconds, then alternates 5 seconds of pump with the valve open and 5 seconds of pump off, 11 phases in total, and then stops.

Descale



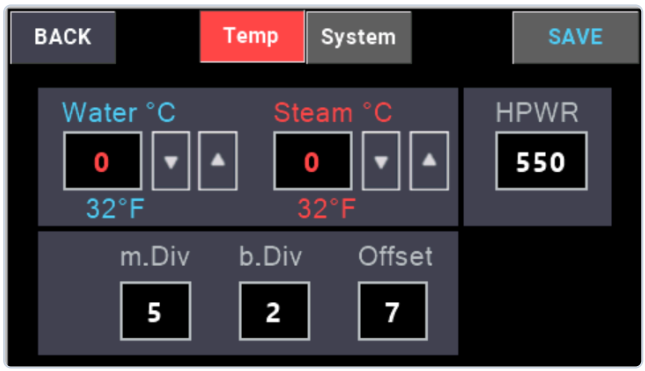
The official manual gives about 45 to 50 minutes. Reading the firmware timing, it works out closer to an hour. Plan on about an hour and do not leave the machine unattended.

1. Fill the tank to MAX with water and descaling solution (the official manual suggests 4 to 6 tablespoons of citric acid per tank).
2. Fit a blind basket and lock the portafilter in.
3. Put a container of 2 litres or more under the steam wand and open the steam valve one full turn.
4. Open the Clean tab, choose Descale, then turn the brew switch on.
5. The machine runs on its own. Check the tank about every 30 minutes.
6. When the solution runs out, empty the container, fill the tank with clean water, and let the program continue. The pump should re-prime itself.
7. When the screen shows FINISHED, turn the brew switch off.
8. Go back to the home screen and select any profile to return to normal brewing.

What the program does: the heater is held off, the valve opens, then it repeats about 30 s of slow pumping, 60 s of rest to let the solution work, and a 10 s burst followed by a pulse sequence with the valve. A progress bar counts 100 steps. Turning the brew switch off at any time stops it.

11. Settings

Temp page



- **Water:** Brew temperature target, saved per profile (93°C, varies by profile).
- **Steam:** Steam temperature target, for the whole machine (155°C).
- **HPWR:** Maximum pulse width of the heater relay (550).
- **M. Div:** Divider for heating when not brewing, used with HPWR (5).
- **B. Div:** Divider for heating during a brew (3).
- **Offset:** Corrects the thermocouple reading to the real water temperature (7°C).

System page



- **LC1 Factor, LC2 Factor:** load-cell calibration values for the two scale cells.
- **Pump Zero:** how much water the pump moves per cycle at zero, in mL. It is used for flow calculation.
- **Warmup:** shows a "Warming Up" message for 15 minutes after power-on.
- **LCD Sleep:** minutes before the screen sleeps (factory 16; 0 disables). This only affects the screen.
- **LCD Brightness:** 0 to 100%.
- **LED Control:** opens the LED page.
- **Reset to defaults:** returns every setting and all five profiles to factory values. Use with care.
- **Build ver. and Nextion ver.:** the firmware build and the display file version, useful when asking for help.

LED page (needs the ToFnLED)



- **Power:** tank LED on or off.
- **Disco:** colour effects during brewing.
- **Red, green, blue sliders:** the LED colour.

Standby Timer (More page)

This is separate from LCD Sleep. It controls the boiler, not the screen.

- **Standby Timer:** on or off.
- **Seconds:** idle time before standby starts (for example 900 is 15 minutes).
- **Idle Temp:** the boiler target while in standby. A low value such as 16°C leaves the heater effectively off.

When the idle time passes, the display saves your temperature, lowers the target to the Idle Temp and goes to the home screen. Touch the screen to wake it, and the normal temperature is restored and the machine heats up again.

Notes for this build: the stock firmware did not act on this setting, so the firmware was changed to watch the display's temperature target directly. Flipping the brew switch does not wake the standby, so touch the screen first or the shot will start with a cold boiler. Also avoid pressing SAVE during standby, because that would store the idle temperature as the profile temperature.

Press SAVE on a settings page to keep any changes.

12. Scales and weight

- **Hardware scales:** two load cells under the drip tray. They tare when a shot starts. In this build, the controller tares before the pump starts and retries if the first attempt fails, which fixes the false "high weight" some shots used to start with.
- **Predictive scales:** used when no hardware scales are fitted. Weight is estimated from pump flow; use Reset weight at the first drops.
- **Stop on weight**, the brew ratio and the Weight above exit conditions work with either type.

13. Messages and what they mean

- **Fill the water tank!:** Tank level is under 10%. Refill.
- **Filling boiler!:** Start-up boiler fill. Wait.
- **Brew Switch ON!:** The brew switch was on at start-up. Turn it off.
- **Releasing pressure!:** The controller is releasing residual pressure above 0.7 bar.
- **TEMP READ ERROR:** The thermocouple gives no valid reading. Pump and heaters are switched off until it recovers. Check wiring. With the steam switch on, the message reads COOLDOWN.
- **TURN STEAM OFF NOW!:** Steam has been on too long. Turn the steam switch off.
- **WATCHDOG RESTARTED:** The controller restarted itself after freezing. If it repeats, report it.
- **FINISHED:** The descale program is complete.
- **Profile loaded! / Profile switched! / Update successful!:** Confirmations after loading, changing or saving a profile or setting.
- **Data out of range!:** A value you entered was rejected. Correct it and save again.

14. Care

Flush the group after shots, back-flush regularly, and descale when the machine needs it. For routine cleaning intervals, follow the official Gaggia guidance for your machine, because this manual does not set any.

15. Troubleshooting

- **No build number on the splash screen:** Display and controller are not communicating. Check the serial wiring and that the controller is running.
- **Shot starts with a high weight reading or stops immediately:** Tare problem. This build tares before the pump starts and retries. If it still happens, check the scale wiring.
- **Boiler stays cool after being idle:** Standby is active. Touch the screen.
- **Boiler never cools or heats at idle:** Check that the Standby Timer is on and the Idle Temp is set.
- **Two profiles look highlighted at start-up:** The display and firmware remembered different profiles. This build now follows the display's profile at start-up. Tap the one you want if it still happens.
- **LED does not light or tank level is blank:** The LED and distance-sensor options are missing from the build settings, or the ToFnLED is not connected. See section 16.
- **Temperature reads wrong:** Adjust the Offset on the Temp page.
- **Scale reads wrong:** Re-check the LC1 and LC2 factors on the System page.

16. Firmware notes for this build

- **Updating:** the firmware is built with PlatformIO in VS Code and flashed with an ST-Link programmer. Use the environment that matches the board (for example the lego-stlink environment).
- **extra_defines.ini:** a file next to platformio.ini holds the options for this machine. Without it, the LED and tank-level sensor do not work. This machine uses:
  ```
  [extra]
  build_flags =
  -DLED_NCP5623
  -DTOF_VL53L0X
  ```
- **Display:** the 4.3" display file is Nextion_43_26JUN2024 (from the Photonsters Gaggiuino_35 project, folder Nextion_43). Keep a copy of the .tft file you flashed.
- **Changes made to this build:** a more reliable tare at shot start (confirmed working), boiler standby driven by the display's Standby Timer (confirmed working), and following the display's remembered profile at start-up (written; not yet confirmed on the machine).
- **Not changed:** the water level LED lockup has not recurred with the original LED code, but its cause was not identified.

17. Quick reference: factory defaults

- **Brew temperature:** 93°C (92, 95, 89, 93 for the other profiles).
- **Steam temperature:** 155°C.
- **Temperature offset:** 7°C.
- **HPWR / M. Div / B. Div:** 550 / 5 / 3.
- **Coffee dose:** 18 g (20 g for Londinium).
- **Brew ratio:** 1:2.
- **LCD Sleep:** 16 minutes.
- **Warmup:** Off.
- **Home On Finish:** Off.
- **Brew Delta:** On.
- **Basket Prefill:** Off.

18. Sources and limits

- Official Gaggiuino Gen 2 User Manual: gaggiuino.github.io/#/learning/user-manual-gen2 (page dated 03/11/2026).
- Gaggiuino project home page feature table (Gen 2 is STM32 with Nextion): gaggiuino.github.io.
- Profile and stop-condition structure: public design discussion on the Zero-bit/gaggiuino GitHub repository.
- This machine's firmware source and 4.3" display project file.

Limits: the official site and GitHub source pages could not be read directly, so the official manual was read from a PDF you supplied. Calibration procedures, electrical installation and Gaggia's own maintenance schedule are not covered here. Behaviour described as coming from the firmware was read from code and has not all been tested on the machine.